$+ \quad \times$

$- \quad \div$

ES: 1

$\mathcal{B}\{|0\rangle, |1\rangle\}$

$$\Delta A \Delta B \geq \frac{1}{2}|\langle [AB] \rangle|$$

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 0 \end{pmatrix} \quad B = \begin{pmatrix} 0 & i \\ -i & 2 \end{pmatrix} \quad |\psi\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

ES.2

$$|\psi\rangle = \sqrt{\frac{2}{3}}\,|x_+\rangle + \frac{1}{\sqrt{3}}|x_-\rangle \qquad \mathcal{B}\{|z_+\rangle, |z_-\rangle\}$$

$$|x_+\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}}|z_+\rangle + \frac{1}{\sqrt{2}}|z_-\rangle$$

$$|x_-\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{\sqrt{2}}|z_+\rangle - \frac{1}{\sqrt{2}}|z_-\rangle$$

$$|\psi\rangle = \sqrt{\frac{2}{3}}\cdot\frac{1}{\sqrt{2}}|z_+\rangle + \sqrt{\frac{2}{3}}\cdot\frac{1}{\sqrt{2}}|z_-\rangle + \frac{1}{\sqrt{3}}\cdot\frac{1}{\sqrt{2}}|z_+\rangle - \frac{1}{\sqrt{3}}\cdot\frac{1}{\sqrt{2}}|z_-\rangle$$

$$= \frac{1}{\sqrt{3}}|z_+\rangle + \frac{1}{\sqrt{6}}|z_+\rangle + \frac{1}{\sqrt{3}}|z_-\rangle - \frac{1}{\sqrt{6}}|z_-\rangle$$

$$= \frac{\overset{2}{\sqrt{2}}+\overset{1}{\sqrt{3}}}{\sqrt{6}}|z_+\rangle + \frac{\sqrt{6}-\sqrt{3}}{\sqrt{6}}|z_-\rangle$$ FIX

EX: 3

$$|\phi\rangle = \frac{1}{2}|0_A\rangle|0_B\rangle + \frac{\sqrt{3}}{2}|0_A\rangle|1_B\rangle \qquad \rho_A = ?$$

$$\rho_A = |\phi\rangle\langle\phi| = \left(\frac{1}{2}\langle 0_A|\langle 0_B| + \frac{\sqrt{3}}{2}\langle 0_A|\langle 1_B|\right)\left(\frac{1}{2}|0_A\rangle|0_B\rangle + \frac{\sqrt{3}}{2}|0_A\rangle|1_B\rangle\right)$$

$$= \frac{1}{4}\langle 0_A|0_A\rangle\langle 0_B|0_B\rangle + \frac{\sqrt{3}}{4}\langle 0_A|0_A\rangle\langle 0_A|1_B\rangle + \frac{\sqrt{3}}{4}\langle 0_A|0_A\rangle\langle 1_B|0_B\rangle + \frac{3}{4}\langle 0_A|0_A\rangle\langle 1_B|1_B\rangle$$

Es. 4

$$M = \begin{pmatrix} 1 & i & 0 \\ -i & 2 & -i \\ 0 & i & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1-\lambda & i & 0 \\ -i & 2-\lambda & -i \\ 0 & i & 1-\lambda \end{pmatrix} = (1-\lambda)\,\mathrm{DET}\begin{pmatrix} 2-\lambda & -i \\ i & 1-\lambda \end{pmatrix} - i\,\mathrm{DET}\begin{pmatrix} -i & -i \\ 0 & 1-\lambda \end{pmatrix} + 0$$

$$= (1-\lambda)\left[(2-\lambda)(1-\lambda)+i^2\right] - i\left[-i+i\lambda\right]$$

$$= (1-\lambda)\left[2-2\lambda-\lambda+\lambda^2-1\right] + i^2 - i^2\lambda$$

$$= (1-\lambda)\left[\lambda^2-3\lambda+1\right] - 1 + \lambda$$

$$= \lambda^2 - 3\lambda + \cancel{1} - \lambda^3 + 3\lambda^2 \cancel{-\lambda} \cancel{-1} + \cancel{\lambda}$$

$$= -\lambda^3 + 4\lambda^2 - 3\lambda = 0$$

$$= \lambda(-\lambda^2 + 4\lambda - 3) = 0 \qquad \lambda = 0$$

$$-\lambda^2 + 4\lambda - 3$$

$$16 + 4(-3) \qquad \Delta = 4$$

$$\frac{-4 \pm 2}{-2} = \begin{cases} 3 \\ 1 \end{cases}$$

Es: 5

$$|\psi(0)\rangle = |x_+\rangle \qquad t = 0$$

$$H = \begin{pmatrix} 2\hbar\Omega & 0 \\ 0 & \hbar\Omega \end{pmatrix}$$

ES: 6 $|\psi\rangle = \sqrt{\frac{2}{3}}|z_+\rangle + \frac{1}{\sqrt{3}}|z_-\rangle$

FIND DENSITY MATRIX

$$|\psi\rangle\langle\psi| = \left(\sqrt{\frac{2}{3}}|z_+\rangle + \frac{1}{\sqrt{3}}|z_-\rangle\right)\left(\sqrt{\frac{2}{3}}\langle z_+| + \frac{1}{\sqrt{3}}\langle z_-|\right)$$

$$= \frac{2}{3}|z_+\rangle\langle z_+| + \frac{\sqrt{3}}{3}|z_+\rangle\langle z_-| + \frac{\sqrt{2}}{3}|z_-\rangle\langle z_+| + \frac{1}{3}|z_-\rangle\langle z_-|$$

$$= \begin{pmatrix} \overset{00}{2/3} & \overset{01}{\sqrt{2}/3} \\ \overset{10}{\sqrt{2}/3} & \overset{11}{1/3} \end{pmatrix} = \rho$$

BLOCH VECTOR

$$x = 2\mathrm{Re}(\rho_{01}) = 2\mathrm{Re}(\sqrt{2}/3) = 2\sqrt{2}/3$$

$$y = 2\mathrm{Im}(\rho_{10}) = 2\mathrm{Re}(\sqrt{2}/3) = 0$$

$$\rho = \frac{1}{2}\begin{pmatrix} 1 + a_z & a_x - i a_y \\ a_x + i a_y & 1 - a_z \end{pmatrix}$$

$$z = \rho_{00} - \rho_{11} = 2/3 - 1/3 = 1/3$$

ES: 7 $\frac{1}{\sqrt{2}}\left(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle\right)$

$\boxed{Es: 8}$

a. $\{\overset{a_1}{|0\rangle+|1\rangle};\ \overset{a_2}{|1\rangle+|2\rangle};\ \overset{a_3}{|2\rangle}\}$

$\cancel{\langle a_1|a_1\rangle = (\langle 0|+\langle 1|)(|0\rangle+|1\rangle) = 2}$ $\quad \langle a_1|a_2\rangle = (\langle 0|+\langle 1|)|(|1\rangle+|2\rangle) = 1$ NO

b. $\overset{b_1}{|0\rangle-|1\rangle}\quad \overset{b_2}{|0\rangle+|1\rangle}\quad \overset{b_3}{|2\rangle}$

$\cancel{(\langle 0|-\langle 1|)(|0\rangle+|1\rangle) = 1-1=0}$ $\quad \langle b_1|b_2\rangle = (\langle 0|-\langle 1|)|(|0\rangle+|1\rangle) = 1-1 = 0$

$\langle b_1|b_3\rangle = (\langle 0|-\langle 1|)|2\rangle = 0$

$\langle b_2|b_3\rangle =$

c $\{\overset{c_1}{2|0\rangle}\quad \overset{c_2}{|1\rangle}\quad \overset{c_3}{i|2\rangle}\}$ $\quad \langle c_1|c_2\rangle = 2\langle 0|1\rangle = 0$

$\cancel{\langle c_1|c_1\rangle = 4\langle 0|0\rangle = 4}$

d. $\overset{d_1}{\frac{|0\rangle-|1\rangle}{\sqrt{2}}}\quad \overset{d_2}{\frac{|0\rangle+|1\rangle}{\sqrt{2}}}\quad \overset{d_3}{|2\rangle}$ $\quad \langle d_1|d_2\rangle = \left(\frac{\langle 0|}{\sqrt{2}}-\frac{\langle 1|}{\sqrt{2}}\right)\left|\left(\frac{|0\rangle}{\sqrt{2}}+\frac{|1\rangle}{\sqrt{2}}\right) = \frac{1}{2}-\frac{1}{2} = 0\right.$

e $\overset{e_1}{|0\rangle}\quad \overset{e_2}{|1\rangle+|2\rangle}\quad \overset{e_3}{|2\rangle}$

$\langle e_1|e_2\rangle = 0$

$\boxed{\text{ES } 9}$

$\boxed{\text{ES } 10}$ $O = 3|0\rangle\langle 0| - |1\rangle\langle 1| + (2+i)|0\rangle\langle 1| + (2-i)|1\rangle\langle 0|$ $\quad |\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$

EXP. $\langle\psi|O|\psi\rangle = \langle\psi|\left(3|0\rangle\langle 0| - |1\rangle\langle 1| + (2+i)|0\rangle\langle 1| + (2-i)|1\rangle\langle 0|\right)|\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$

$= \langle\psi|\left(\frac{3}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle - \frac{(2+i)}{\sqrt{2}}|0\rangle + \frac{(2-i)}{\sqrt{2}}|1\rangle\right.$

$= \frac{1}{\sqrt{2}}\langle 0| - \frac{1}{\sqrt{2}}\langle 1| \, \Big| \, \frac{3}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle - \frac{(2+i)}{\sqrt{2}}|0\rangle + \frac{(2-i)}{\sqrt{2}}|1\rangle$

$3 - \frac{2+i}{2} - \frac{1}{2} - \frac{2-i}{2}$

$= 3 - \frac{1}{2} - 2 - i$

$= \frac{1}{2} - i$

$\langle \psi | o^2 | \psi \rangle$

$\langle \psi |$